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Independent Study (CSC630) Project Description
Sparse Tensor-Network Based Quantum Simulation Optimization

As a graduate student in computer science, I aspire to take up an independent study course (CSC 630) in the upcoming semester under the guidance of Dr. Frank Mueller. Our goal is to develop a novel numerical library that can efficiently store matrices or tensors and perform contraction faster by exploiting sparsity patterns specific to quantum circuits but also applicable to machine learning.

This project is part of a wider collaborative effort that involves several universities (NSF QLCI) and aims to optimize tensor network contraction at different levels. These levels include proposing memory hierarchy and microarchitecture-aware representations, efficient data (re-)arrangement, designing memory hierarchy-aware and balanced algorithms with smart page arrangement, and addressing the curse of dimensionality through memoization and intelligent data allocation. The ultimate objective of this collaborative effort is to speed up six application scenarios by leveraging scalable and highly optimized sparse tensor networks on distributed heterogeneous systems.

Under the guidance of Dr. Frank Mueller, I aim to investigate the impact and extent of sparsity in quantum simulations using various techniques such as state vector simulation and tensor network simulation (TTN, PEPS, MERA, etc.). The goal of this independent study is to develop an optimized and high-performance parallel C++ numerical library that can efficiently store matrices or tensors and perform contraction faster by exploiting sparsity patterns. We would then focus on comparing its performance against modern numerical libraries like Numpy, Pytorch, Jax, etc. for quantum simulation benchmarks, and analyze the impact of sparsity on the performance of the proposed library.

Additionally, We aim to interface the library with Python APIs and utilize it with quantum software such as quimb (Google's cirq) and KOALA (Cyclops group at UIUC). The project will also target improving simulation performance using dynamic programming techniques and other methods.

This work will contribute to the development of optimized and efficient numerical libraries that can be used by researchers and practitioners to accelerate quantum simulations and machine learning applications. The library could potentially be used in a wide range of fields, from quantum computing to material science, and could provide significant benefits to industry and academia alike.

In conclusion, this independent study project represents an exciting opportunity for me to contribute to the ongoing efforts to optimize tensor network contraction and develop a novel numerical library for quantum simulations and machine learning. The project is ambitious and will require a significant amount of effort and expertise, but the potential benefits are significant, and the results could have a lasting impact on the field.